

Bellabeat Consumer Insights

Executive Summary: Smart Tracker Behavior & Marketing Opportunities

Prepared By: Narayani SS
Focus: Digital Wellness Strategy

Strategic Context: Bellabeat designs health-focused technology for women. By analyzing public smart-device activity records across multiple telemetry tables, this study identifies usage patterns across steps, intensities, and day-of-week trends to shape Bellabeat's user acquisition and app engagement strategy.

88%

HIGHLY ACTIVE USERS

29 of 33 logged >20 days/month

8,153

PEAK DAILY STEPS (SAT)

Vs. 6,933 minimum on Sun

12PM & 5–7PM

PEAK INTENSITY WINDOWS

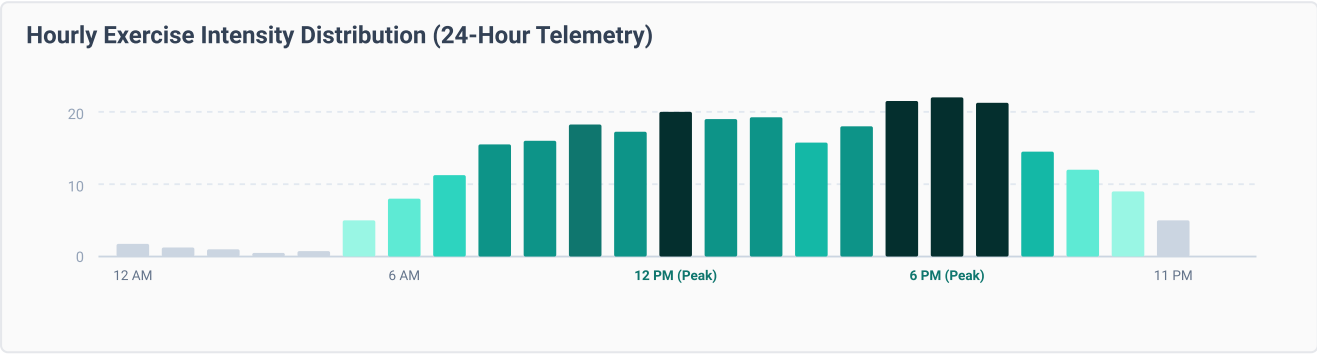
Lunch & post-work exercise spikes

DATA PIPELINE & INTEGRITY VALIDATION

Data hygiene was conducted in Google Sheets and SQL (Google BigQuery). Relational integrity checks confirmed that `DailyCalories`, `DailyIntensities`, and `DailySteps` perfectly matched the master `DailyActivity` table across 940 distinct user-date observations using full outer joins.

DAY-OF-WEEK ACTIVITY & EXPENDITURE BASELINE

DAY OF WEEK	TOTAL AGGREGATED STEPS	AVG STEPS / DAY	AVG DISTANCE (KM)	AVG CALORIES BURNT
Saturday	1,010,969	8,153	5.85	2,355
Tuesday	1,235,001	8,125	5.83	2,356
Monday	933,704	7,781	5.55	2,324
Wednesday	1,133,906	7,559	5.49	2,303
Friday	938,477	7,448	5.31	2,332
Thursday	1,088,658	7,406	5.31	2,200
Sunday	838,921	6,933	5.03	2,263



SYNTHESIS OF CORE ANALYTICAL FINDINGS

The data reveals distinct behavioural patterns that directly inform how Bellabeat can personalize user nudges, schedule marketing campaigns, and expand female health tracking features.

1. Step & Calorie Positive Correlation

Tableau bivariate analysis confirmed a strong positive relationship between daily steps and total calorie burn. High-intensity exercise yields direct energy expenditure improvements.

2. The "Sunday Drop-off" Phenomenon

Sunday step averages decline by 15.0% compared to Saturday peaks (6,933 vs. 8,153 steps), identifying a systemic weekly slump across tracked users.

3. Dual-Peak Commute/Lunch Activity

Active intensity surges precisely between 12:00 PM (lunch break) and 5:00 PM – 7:00 PM (post-work commute and workout windows).

4. Data Gap: Menstrual Tracking Absence

Standard public fitness tracker data completely omitted hormonal and menstrual cycle metrics—representing Bellabeat's primary competitive differentiator.

ACTIONABLE RECOMMENDATIONS FOR BELLABEAT

1. Behavioral Push Notifications Synced to Peak Windows FEATURE UPDATE

Automate app reminders calibrated to user routine spikes: deliver a gentle mobility prompt at **12:00 PM** and post-work fitness class recommendations between **5:00 PM and 6:30 PM** to capitalize on high-intent time slots.

2. Launch the "Sunday Wellness Reset" Campaign MARKETING STRATEGY

Address the Sunday activity deficit by rebranding Sundays around restorative wellness: offer guided mindfulness, sleep preparation routines, light walking challenges, or partner product discounts to maintain weekly habit consistency.

3. Goal Gamification & Tiered Milestone Rewards ENGAGEMENT & RETENTION

Leverage the strong step-to-calorie correlation by establishing achievable milestones (e.g., 7,500 daily steps). Reward sustained consistency with redeemable Bellabeat membership points, apparel discounts, or wellness partner gift cards.

4. Integrate Cycle-Aware Workout Management VALUE PROPOSITION

Capitalize on Bellabeat's core female wellness identity by pairing daily activity tracking with menstrual cycle phases (follicular, luteal). Offer adaptive fitness and recovery recommendations based on energy changes across the monthly cycle.

Methodological Note: Derived from 30+ Fitbit user logs via Kaggle (Amazon Mechanical Turk, April–May 2016). Validated via BigQuery SQL aggregation and Tableau exploratory visual analytics.